Section-5

SOLID CORE INSULATORS

1 SCOPE: Refer Clause-1 of Section-I, General

This specification covers design, manufacture, testing before despatch, packing, forwarding and delivery, Free at Substation Site of 220kV and 132kV Solid Core Insulators required for 220kV and 132kV Sub-Stations. One 220kV Solid Core Insulator shall consists of two insulator units forming a stack.

2 STANDARDS:

The Solid core type insulator shall be in accordance with ISS-5350-II latest issue. Other standards applicable are as here under.

IS:2544	:	Porcelain Post Insulator for systems with nominal voltage greater than 1000V.
IS:2629	:	Recommenced practice for hot dip galvanised of Iron and Steel.
IS: 2633	:	Method for testing uniformity of coating of Zinc coated articles.
IS:6735	:	Fasteners-Spring lock washers for screws with cylindrical heads.
IS:2016	:	Plain Washer
IEC 71	:	Electrical Clearances.

3 MATERIAL

The porcelain part shall be permanently secured in a metal case to be mounted on the supporting structures. They shall be made up of interchangeable units and shall be capable of being mounted on the supporting structures. Suitable bolts, washers required for inter connecting the solid core insulator units to form a 220kV Solid Core insulator stack shall be supplied.

The porcelain shall be sound and homogenous free from defect, laminations and other flaws or imperfections which might effect the mechanical or dielectric quality. These should be thoroughly verified tough, impervious to moisture and smoothly glazed. The finished insulator shall be brown in colour and the glaze shall be uniform and free from blisters, burns and similar other defects. The metal fittings shall be cemented properly to the insulator.

The metal fittings for the insulators should be made from high grade cast iron or malleable steel casting and all the ferrous metal parts shall be hot dip galvanised. The coating or metal parts shall withstand minimum four one minute dips in copper sulphate solution as per IEC-168. Cementing between metal to porcelain should be carried out by high grade Portland cement with special care to check the expansion and contraction problem. The metal, cement and porcelain joints shall be painted with suitable paint to offer protection, against different thermal expansion. The porcelain and metal parts shall be assembled in such a manner that any thermal expansion differential between the metal and porcelain parts throughout the range

of temperature variation shall not loosen the parts or create undue internal stresses which may affect the electrical or mechanical strength and rigidity. The cap and base of the insulator shall interchangeable with each other. These shall confirm in all respects to IS: 5359 (Part-II) latest issue. The insulators unit shall be assembled in a suitable jig to ensure correct positioning of the top and bottom metal fittings relative to one another. The faces of the metal fittings shall be parallel and at right angles to the axis of the insulator and the corresponding holes in top and bottom metal fittings shall be in a vertical plane containing the axis of the insulator.

4 DESIGN:

The insulators shall be designed to withstand any shocks to which they may be subjected by the operation of any of the associated equipment. The design shall be such that stresses due to expansion and contraction in any part of the insulator shall not lead to breakage.

5 GENERAL REQUIREMENTS:

The insulators are intended for erection with 220kV and 132kV Isolators and for support purpose in the APTRANSCO EHT Sub-stations. The switch portion of the Isolators are being procured by the APTRANSCO separately. The particulars of the system, the insulation levels to be withstood by the insulators and other particulars are indicated below:

		The values under column shall correspond to their usage at 220kV and 132kV insulators.	
Nominal system voltage	:	220kV (Rms)	132kV (Rms)
Highest System voltage	:	245kV (Rms)	145kV (Rms)
Standard impulse withstand Voltage wet	:	1050 kV (Peak)	650kV (Peak)
One minute Power Frequency Withstand voltage.	:	460kV (Rms)	275kV(Rms)
No. of insulator units per stack:		Two	One
Height of each insulator unit		1150 mm	1500 $\pm$ 2.5
Total height of stack		2300 $\pm$ 3.5mm	
Pitch circle diameter		4 Nos. of tapped holes of size M16 on the top and 4 Nos. of plain Holes suitable for bolts of size 18mm dia on the bottom.	
Top		127 mm	127mm
Bottom		225mm	184 mm
Total creepage distance (min)		7595mm	4495 mm
Mechanical strength required(Min)			
Cantilever		6000N	6000N
Tensile		110000 N	80000 N

Torsion	4500 N-M	4500 N-M
Compression	220000 N	160000 N

All other mechanical and electrical features shall be in conformity with port insulator designation C6-1050, C6-650 of IS.5350 Part-II latest issue.

Note: Tolerance of dimension shall be as per clause No.9.7.2 of IS 2544.

6 TYPE TESTS:

The Solid Core insulator of the type design offered shall have been fully type tested by the bidder as per the relevant standard. The type tests must have been conducted not earlier than five years form the date of bid opening. The Bidder shall furnish one set of type test reports along with offer.

7. ACCEPTANCE TESTS:

All acceptance and routine tests as stipulated in the relevant standards shall be carried out by the Supplier in the presence of Purchaser's representative.

Acceptance test whenever possible shall be conducted on the complete unit. The number of samples to be subjected to acceptance tests shall be decided by the Purchaser at the time of actual testing.

8 INSPECTION

The Purchaser's representative shall, at all times, be entitled to have access to the works and all places of manufacture where cable shall be manufactured and the representative shall have full facilities for unrestricted inspection of the Bidder's works, raw materials and process of manufacture and conducting necessary tests as detailed herein.

The supplier shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of cable in its various stages so that arrangements can be made for inspection.

The supplier shall give 15 days (for local supplies) 30 days (for foreign supplies) advance intimation to enable the purchaser to depute his representative for witnessing acceptance and routine tests. All the charges in connection with inspection by purchaser personnel such as travel, accommodation and incidentals shall be borne by the bidder.

No material shall be despatched from its point of manufacture unless the material has been satisfactorily inspected, tested and cleared for despatch.

The acceptance of any quantity of material shall in no way relieve the Bidder of any of his responsibilities for meeting all requirements of the specification, and shall not prevent subsequent rejection if such material is later found to be defective.

9 QUALITY ASSURANCE PLAN : Refer Clause-3 of Section-I, General

10. DOCUMENTATION:

10.1 All drawings shall conform to International Standards Organization (ISO) "A" series of drawing sheets/Indian Standards Specifications. All drawings shall be in ink and suitable for microfilming. All dimensions and data shall be in S.I. Units.

List of drawings and documents:

The bidder shall furnish four sets of following drawings along with his offer.

i) General outline drawings of the complete solid core insulator with technical parameters.

10.2 The Supplier shall within 1 week of placement of order, submit four sets of final versions of all the above said drawings for Purchaser's approval. The Purchaser shall communicate his comments/approval on the drawings to the Supplier with in two weeks of receipt of drawings. The Supplier shall, if necessary, modify the drawings and resubmit four copies of the modified drawings for Purchaser's approval within two weeks from the date of Purchaser's comments. The Purchaser shall communicate approval with in two weeks of receipt of modified drawings. After receipt of Purchaser's approval, the Supplier shall within three weeks, submit 20 prints and two good quality reproducible of the approved drawings for Purchaser's use.

Approval of drawings by Purchaser shall not relieve the Supplier of his responsibility and liability for ensuring correctness and correct interpretation of the latest revision of applicable standards, rules and codes of practices.

11 TEST REPORTS:

i) Four copies of acceptance test reports shall be furnished to the purchaser. One copy will be returned, duly certified by the purchaser and only thereafter shall the materials be despatched.

ii) All records of routine test reports shall be maintained by the Bidder at his works for periodic inspection by the purchaser.

iii) All test reports of tests conducted during manufacture shall be maintained by the Bidder. These shall be produced for verification as and when requested for by the purchaser.

12 PACKING AND FORWARDING & UNLOADING:

Refer Clause-4 of Section-I, General

13 QUANTITY AND DELIVERY REQUIREMENTS : Refer Clause-5 of Section-I, General

14 SUPERVISION SERVICES : Refer Clause-6 of Section-I, General

15 GUARANTEED TECHNICAL PARTICULARS:

The Guaranteed Technical particulars for the equipment being supplied shall be provided with the Bid as specified in this technical specification at Annexure-I**.** The Bids received without guaranteed technical particulars shall be treated as Non-responsive.

ANNEXURE-I

220kV and 132kV SOLID CORE INSULATORS

GUARANTEED TECHNICAL PARTICULARS

(To be filled by the Bidder)

1	Manufacturer	
2	Standard	
3	Type	
4	Insulating material	
5	No.of units per stack	
6	Diameter of the largest shed(mm)	
7.a)	Height of insulator stack mm	
b)	Height of each basic unit mm	
8	Weight kgs (Approx)	
9	Total creepage distance(minimum)mm	
10	Total creepage distance mm	
11	Flashover Voltage	
	Dry kV	
	Wet kV	
	Impulse kV	
12	Withstand test voltage	
	Dry Kvp	
	Wet kVP	
	Impulse kV	
13	Power frequency puncture withstand voltage kV	
14	Working Voltage kV	
15	Mechanical strength	
	a) Cant lever strength	
	b) Tensile strength	
	c) Torsion Strength	
	d) Compression Strength	
16	Pitch cycle diameter	
	Top mm	
	Bottom mm	
17	Drawing No. Reference Drg.	

NOTE:

1. Technical brouchers Literature and drawings to illustrate fully the insulators quoted to be furnished along with tender.

2. 220kV insulators shall be supplied with having such as G.I. Bolts, washers and nuts etc., to form stack.